

DiaTool-DPO: Enhancing Tool-Augmented LLM Dialogue

via Direct Preference Optimization

A model DPO-based method with in-store LLM and automatic paired trajectories for robust slot-filling and tool-call rejection

Audience: ML researchers and NLP engineers | Tone: technical, analytical, concise

Multi-turn RLHF for TA-LLMs

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Failure modes in tool-augmented LLM dialogue control

Core Problem

- Failing to ask follow-up questions when required slots are missing (incomplete user queries).
- Hallucinated tool invocations with ungrounded or fabricated slot values.
- Failing to reject inappropriate or out-of-scope tool calls (unsafe tool usage).

Why Preference Optimization?

SFT reproduces demonstrations but weakly encodes multi-turn decision quality under uncertainty.

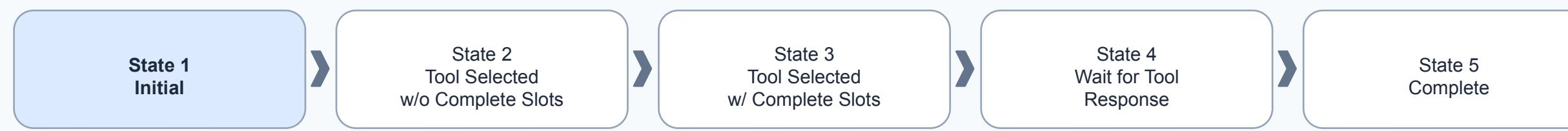
DiaTool-DPO introduces trajectory-level preferences to reward:

- Clarification before calling tools
- Correct slot completion
- Appropriate tool-call rejection

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

State machine foundation for trajectory-level preference optimization

Reference: figure_1.jpg (state diagram)



Interpretation The MDP abstracts TA-LLM dialogue into transitions between slot-completeness, tool-execution wait, and completion. This supports systematic generation of chosen/rejected trajectories for DPO training.

Three Query Types Define Distinct State Trajectories

Compact mapping from query class to MDP path and behavior

Query Type	State Trajectory	Brief Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	User provides sufficient slots; model calls tool directly and completes.
Type 2	$1 \rightarrow (2*N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Missing slots trigger N clarification turns before valid tool call.
Type 3	$1 \rightarrow 1$	Out-of-scope request; model rejects tool call and stays at initial state.

Automatic Paired Trajectory Construction Without Human Labels

Chosen vs rejected trajectory pairing by query type and failure pattern

Pair Construction Logic

- Type 1: chosen follows 1→3→4→5; rejected includes unnecessary slot-filling or incorrect transitions.
- Type 2: chosen collects missing slots before tool call; rejected hallucinates slots or calls tool early.
- Type 3: chosen rejects tool call and remains at State 1; rejected incorrectly invokes tool.

Dataset Composition

Type	Count	Difficulty
Type 1	6,286	Low
Type 2	8,432	High
Type 3	4,214	Medium
Total	18,932	Mixed

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Benchmark comparison across Prop.-8B, Prop.-3.1B, and LLaMA-3-8B

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.514	0.638	0.611	0.522	0.571	0.571
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Before/after impact when adding DiaTool-DPO to SFT



Slot
0.639 → 0.917
(+43.5%)

Relevance
0.826 → 0.913
(+10.5%)

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

Implications for deploying aligned tool-use dialogue systems

- MDP formulation (5 states, 3 query types) enables systematic chosen/rejected trajectory construction.
- Automatic preference pair generation removes dependence on human preference labeling workflows.
- DiaTool-DPO complements SFT: DPO-only underperforms, while SFT + DiaTool-DPO yields best Macro Avg.
- Largest gains appear in slot-filling robustness and appropriate tool-call rejection behavior.
- Method is language-agnostic (Korean→English transfer), supporting production TA-LLM deployment at smaller model scales.

Outcome: Near GPT-4o-level conversational tool-use quality on smaller open-source models.